SAL EDUCATION CAMPUS

SAL COLLEGE OF ENGINEERING (113)

CE,IT,CSE Department

SUBJECT: Computer Organization &Architecture

SUBJECT CODE: BE04000251

SEMESTER- IV

Mid Semester Syllabus Summer-2026

Unit No.	Content
1	Structure of Computers: Computer types, Functional units, Basic operational concepts, Von Neumann Architecture, Bus Structures, Software, Performance, Multiprocessors and Multicomputer, Data representation, Fixed and Floating point, Error detection and correction codes.
2	Register Transfer and Micro-operations: Register Transfer language, Register Transfer, Bus and Memory Transfers (Tree-State Bus Buffers, Memory Transfer), Arithmetic Micro Operations, Logic Micro-Operations, Shift Micro-Operations, Arithmetic logical shift unit
3	Basic Computer Organization and Design Instruction codes, Computer registers, computer instructions, Timing and Control, Instruction cycle, Memory-Reference Instructions, Input-output and interrupt, Complete computer description, Design of Basic computer, Design of Accumulator Unit.
4	Assembly Language Programming Introduction, Machine Language, Assembly Language Programming: Arithmetic and logic operations, looping constructs, Subroutines, I-O Programming.
5	Micro programmed Control Organization: Control Memory, Address sequencing, Micro program example, Design of Control Unit
6	Central Processing Unit Introduction, General Register Organization, Stack Organization, Instruction format, Addressing Modes, Data transfer and manipulation, Program control, Reduced Instruction Set Computer (RISC) & Complex Instruction Set Computer (CISC)

Prof.Sweta Shilu

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